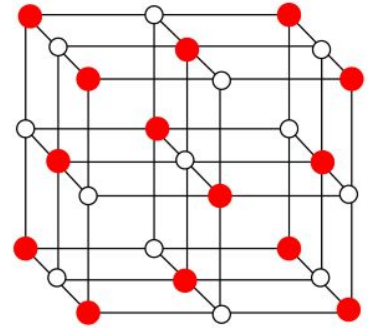


X-Ray Diffraction (XRD)

What is XRD?

XRD is used to determine information about a material's crystal structure and composition.

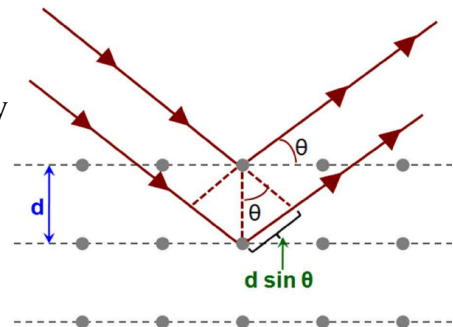
1. X-rays with a specific wavelength (often 0.154 nm) are shot at a sample.
2. The X-rays are scattered by atoms.
3. X-rays, which scatter constructively, are measured and plotted as an intensity versus 2θ graph.



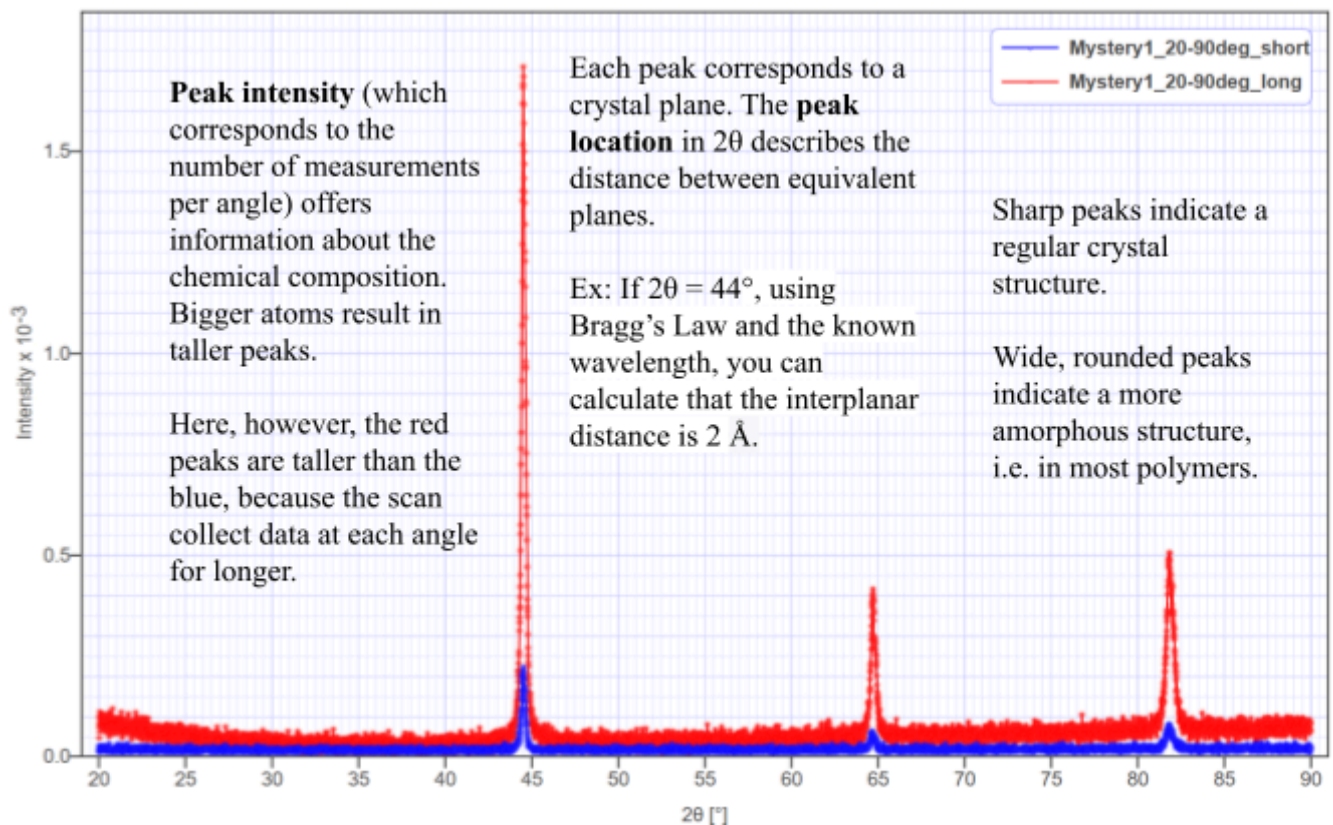
What determines if the X-rays are scattered constructively?

Constructive interference occurs when **Bragg's Law** ($\lambda n = 2d \sin(\theta)$) is satisfied, meaning that angle between the plane and the X-ray (θ) results in the X-ray traveling an integer multiple (n) of the X-ray wavelength (λ).

Essentially constructive interference occurs when there are regularly spaced planes. The angle of the incident X-ray tells us the distance between these planes.



How do you analyze the XRD output?



Instructions on using the Breakerspace X-Ray Diffractometer

The Breakerspace has Panalytical Aeris Research XRD, which is capable of analysing solids, powders, and thin films.

Sample Preparation:

- Sample holders and clay to hold solid samples can be found in the black cabinet near the sink.
- Samples can be leveled with one of two sample leveling presses
- Prepare your sample externally at the sample prep table



Instrument startup:

- Ensure at least one position in the sample changer is empty
- Turn the mains power switch at the rear of the instrument
- Make sure that the cover is closed
- Press the power button to switch on the instrument
- Turn the HT key-switch to switch on the HT generator

Operation:

- Remove plastic sample case cover and put the sample holder in a loading position in the sample case. Replace plastic cover.
- Select a measurement program from the drop-down list
- Put in a descriptive sample name
- Start the measurement
- Export results to the networked workstation or USB drive.
 - The computer's username is xrd and the password is xrd-password. Data can be found in the folder C:\XRD\XRD data. The analysis program is called HighScore Plus.
- Fill out your information in the X-ray safety log book
- Repeat as needed - additional samples can be loaded and added to the queue while current sample is being measured

Instrument shut down:

- Press the Power button to switch off the instrument
- Turn the HT Keyswitch counter-clockwise to switch off the HT generator
- If the instrument will be switched off for a long period of time, you can also switch off the mains power supply

See the Breakerspace website for the extended SOP! Each workstation also has a desktop icon linking to its SOP.

As always, if you need help, contact a lab assistant!